QoS-Aware Sensor Virtualization for Provisioning Green Sensors-as-a-Service

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***Abstract*—This work addresses the problem of efficient virtual sensor formation in sensor-cloud for provisioning green Internet of Things (IoT)-based services. The sustainability of sensor-cloud primarily depends on its quality of service (QoS), resource utilization efficiency, and the revenue earned by its stakeholders. As sensor nodes are resource-constrained, it is important to minimize the overall energy consumption at these devices and for the overall network, thereby reducing the cost of service provisioning. Additionally, it is important to ensure that there is no compromise in the QoS to establish a continuous inflow of revenue for the service-provider. Existing works on virtual sensor formation in sensor-cloud barely consider resource consumption due to multihop communication from source nodes to the base-station, which plays a significant role in determining the resource consumption of the network. Thus, in this work, we model this problem using a game-theoretic approach and propose a hybrid algorithm, named GS2, to solve it. We observe that, by modifying the price paid to the sensor-owners, the service-provider is able to control the resource utilization of the network. Simulations show that GS2 reduces the overall energy consumption by 15.92-24.18%, and increases the revenues of the service-providers and sensor-owners by 8.33-18.33% and 21.05-38.09%, respectively.**

***Index Terms*—Sensor-cloud, IoT services, virtual sensor, sensor networks, game theory, green communication.**

I. Introduction

Sensor-Cloud is a novel paradigm which aims to unify two widely used technologies – Wireless Sensor Networks (WSN) [1] and Cloud, and bring their combined advantages to a single platform. It gives rise to a new form of Internet of Things (IoT)-based cloud service, termed as Sensors-as-a-Service (Se-aaS), in which virtualized instances of physical sensor nodes are provisioned by a sensor-cloud service provider (SCSP) to users in the form of service units [2]. SCSP traditionally obtains these sensor nodes on rental basis from the sensor-owners who are responsible for purchasing, deploying, and maintaining these nodes. Similar to traditional cloud services, Se-aaS follows the pay-per-use model in which users only pay to the SCSP for the units of service consumed by them. Thus, the sensor-cloud architecture offers several advantages. Firstly, due to virtualization, the same physical sensor nodes are used to serve multiple end-user applications, thereby leading to increased resource utilization efficiency. Secondly, the end-users are able to just use these services without being concerned about the backend functionalities. Thirdly, sensor-cloud is economically beneficial to the involved entities as the users only pay as long as they consume Se-aaS, while the SCSP and the sensor-owners also get an opportunity to earn revenue. Because of its many advantages, sensor-cloud has received considerable research attention in the last few years. Clearly, in sensor-cloud, the revenue earned by the SCSP and the sensor-owners depends on the quality-of-service (QoS) provided to the users and the efficient utilization of the resource-constrained sensor nodes. Thus, in this work, we design an optimal sensor virtualization scheme for provisioning Se-aaS, while ensuring high QoS in terms of delay and service availability, and high resource utilization in terms of energy and bandwidth capacity of the sensor nodes.

A. Motivation

Sensor-cloud is a large-scale infrastructure comprising a huge number of energy-constrained edge devices, viz., sensor nodes and gateways. It is capable of supporting a huge number of services simultaneously and hence, demands a huge amount of energy. Thus, to ensure the sustainability of sensor-cloud, it is important to ensure the the edge devices in an energy-efficient manner to ensure the greenness of its design. Moreover, as discussed earlier, in sensor-cloud, it must also be ensured that the QoS requirements of the users are fulfilled to ensure their satisfaction. For maintaining QoS, the SCSP must ensure that the sensor nodes have sufficient resources to serve the requests assigned to it. In other words, the assigned source nodes must have sufficient lifetime and storage and processing capacities to prevent abrupt service disruptions. Additionally, since the sensor nodes utilize multi-hop communication for transmitting data, the energy consumption of the intermediate nodes must also be considered, while allocating services to the nodes to prevent unwanted delays due to change of routes. On the other hand, it is equally important to maintain the profits of the different stakeholders, viz., SCSP and sensor-owners, of sensor-cloud to maintain their participation in the Se-aaS market. The revenues earned by these entities depend on the strategic utilization of the nodes. Thus, to ensure non-zero profit, the SCSP must try to serve more number of services using the same set of sensor nodes, while taking into account their capacities. Whereas, the sensor-owners must choose the sensor nodes for participation in provisioning a particular service, only if the cost of maintaining the nodes is significantly less than the price received from the SCSP for the usage of the nodes [3].

In the existing literature, researchers proposed a few schemes viz., [4]–[7], for efficient virtual sensor formation

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(VSF), i.e., the allocation of service requests to sensor nodes, in sensor-cloud. However, these schemes primarily focus on the selection of the source nodes for service provisioning and barely consider the energy consumption of the intermediate hop nodes, which plays a significant role in deciding the network lifetime. Although a few schemes, viz., [5], [8], consider the selection of optimal hops for energy efficient data transmissions, these works either do not consider the possibility of serving multiple user applications using the same nodes [5], or barely consider the economic aspects of sensor-cloud [8]. However, these are important and unique aspects of sensor-cloud as the user applications may have varied QoS requirements and service provisioning cost, which need to be considered while forming the virtual sensors (VSs). In this work, we aim to address this lacuna of the existing literature and design an energy efficient QoS-aware sensor virtualization scheme for architecting a green sensor-cloud.

B. Contribution

In this work, we propose GS2, an optimal VSF scheme for green sensor-cloud, while considering the QoS requirements of the services, resource consumption of the sensor nodes, and revenues earned by the SCSP and sensor-owners. The primary contributions of this work are:

1. We model the scenario of energy- and cost-efficient VSF in sensor-cloud using a game-theoretic approach in which rational sensor-owners cooperate with each other and the SCSP to decide the set of sensor nodes to be used to form VSs.
2. Based on the aforementioned model, we design a distributed scheme using coalition formation game to decide the optimal data transmission path for each possible source node, while minimizing the net energy consumption and maximizing the profits of the sensor-owners.
3. We design a centralized scheme for the SCSP to decide the optimal set of source and corresponding hop nodes to form the set of VSs, while ensuring a trade-off between its own profit and the global energy consumption of the network.
4. We design a simulation environment for sensor-cloud to analyze the performance of the proposed scheme and compare it to the existing benchmarks in terms of energy consumption, QoS, and economic benefits.

The paper is organized as follows. Section II discusses some of the related literature on sensor-cloud. Section III presents the system model followed by the detailed discussion on the proposed scheme in Section IV. Section V discusses the results of the experiments and Section VI concludes the paper.

II. Related Works

In the last decade, several researchers proposed various schemes for enabling efficient provisioning of IoT-based services [9]–[11]. We discuss some of these works, while dividing them into two categories – (a) resource allocation schemes for IoT services, and (b) VS composition in sensor-cloud.

Resource Allocation Schemes for IoT: Thoma *et al.* [12] studied the different types of IoT services and their corresponding architecture along with a discussion on the business and enterprise integration aspects. Al-Fuqaha *et al.* [13] proposed a rule-based intelligent gateway which unifies IoT services using different protocols for communication and developed a generic IoT service protocol. Kantelis *et al.* [14] proposed a scheme for optimal resource allocation in healthcare IoT. Abdelwahab *et al.* [15] proposed a resource discovery and virtual sensor network formation protocol for IoT devices for provisioning sensing services in a cloud of things architecture. On the other hand, Drosatos *et al.* [16] surveyed the privacy issues in the conext of Industrial IoT. Several works focused on fog [17] and edge-based architectures for IoT services. Cheng *et al.* [18] proposed FogFlow, a programming model with standard interfaces to support heterogeneous and elastic IoT services over the Fog networks. Alameddine *et al.* [19] proposed an optimal task offloading and scheduling scheme for executing IoT services in resource constrained mobile edge networks while considering the heterogeneity of requirements of the tasks and the device capacities. An optimal IoT service placement and load balancing scheme was proposed by Maia *et al.* [20] in which the authors aimed to reduce the QoS violations while considering the resource limitations of the edge devices. Amadeo *et al.* [21] proposed a Named Data Networking-based distributed computing paradigm in which the forwarding mechanisms are designed in such a manner as to reduce the amount of data crossing the network and to allocate services to nodes with available resources. Li *et al.* [22] studied the resource allocation in non-orthogonal multiple access and emphasised on the advantage of buffer technique. In another work, Nguyen *et al.* [23] addressed the problem of Virtual Network Function (VNF) placement and chaining while minimizing the bandwidth consumption and delay of provisioning IoT services using those VNFs. A profit sharing scheme for fog devices providing IoT services was proposed by Ridhawi *et al.* [24] in which the authors discussed the design of a learning-based approach for building workflows to maximise user satisfaction and profit gains of each service provider. Spathi *et al.* [25] evaluated the performance of IoT networks. Abkenar and Jamalipour [26] proposed a protocol for energy efficient association of IoT nodes with Fog nodes to minimize the task delays. Meanwhile, Yao and Ansari [27] discussed an online, reinforcement learning-based algorithm to distribute workload in Fog-based IoT networks fanning in the uncertainty of the communication channels and the nature of future traffic load. In another work, Nguyen *et al.* [28] addressed the problem of energy efficient routing in energy harvesting IoT networks and proposed EHARA, a routing algorithm which considers network QoS and availability of energy sources. Boursianis *et al.* [29] designed a energy harvesting-enabled smart IoT systems for agriculture. In the context of WSNs, Al-Tous and Barhumi [30] proposed a differential game-based scheme in which sensor nodes regulate their transmission energy and data size to reduce energy consumption and distribute traffic load.

VS Composition Schemes for sensor-cloud: Several researchers explored VS composition in sensor-cloud. Chatterjee *et al.* [4] proposed two schemes – COV-I and COV-II – for optimal composition of VSs over the same and different geographic regions. Misra *et al.* [7] proposed a scheme for optimal mapping of physical sensors to VSs using a game the-

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oretic approach, while considering the optimal distribution of service load among multiple sensor nodes and owners and the associated economic benefits. In another work, Chakraborty *et al.* [2] proposed a pricing scheme for sensor-cloud in which the services are allocated to the nodes while considering their physical parameters and trust values. Another trust evaluation scheme for sensor cloud in the context of industrial IoT was proposed by Wang *et al.* [31] in which the authors used mobile edge devices to collect evidence for generating trust values based on a probabilistic graph model and used this value to choose the shortest paths. Delgado *et al.* [8] attempted to increase the service availability and hence the revenue of the SCSP by proposing an optimal resource allocation scheme in which each task is divided into independent subtasks before allocation. Kim [5] and Santos *et al.* [6] proposed resource allocation schemes for sensor-cloud, while considering malicious node behavior and minimizing resource consumption.

Synthesis: From literature survey, we observe the following:

1) The resource allocation schemes for IoT services do not ensure QoS and reduced service provisioning cost, simultaneously. These also do not consider multihop communication which is typical in WSN-based sensor-cloud.
2) Although the resource allocation schemes for sensor-cloud consider service provisioning cost, QoS, and multi-hop communication, these focus mainly on optimal VSF using the source nodes.
3) None of the existing works consider the resource consumption of the intermediate hop nodes. As sensor nodes have limited battery and bandwidth, the energy and bandwidth consumption due to data forwarding and the energy consumption due to state transition impact the network lifetime significantly and hence, must be considered for efficient virtualization.

These necessitate the design of the optimal VSF scheme, GS2, which distributes service load efficiently among the source and intermediate hop nodes, while ensuring reduced energy consumption and high QoS and revenue for the sensor-owners and SCSP.

III. System Model

We consider a single SCSP which provides Se-aaS over an extensive geographical region $\mathcal{R}$, as shown in Figure 1. The SCSP utilizes the sensor nodes deployed by multiple sensor-owners over $\mathcal{R}$. The set of registered sensor-owners is denoted by $\mathcal{O} = \{o_1, o_2, \cdots, o_j\}$, and the set of nodes belonging to each owner o_j is denoted by $\mathcal{N}_j = \{n_1^j, n_2^j, \cdots, n_k^j\}$. We consider that $|\mathcal{N}_j| = K_j$, where K_j denotes the total number of nodes owned by owner o_j. The sensor nodes are heterogeneous in terms of the types of equipped sensors, energy, and storage and computation capabilities. We denote the types of sensors equipped in a sensor node n_k using $T_k = \{t_1^k, t_2^k, \cdots, t_l^k\}$, where $|T_k| = L_k$ and L_k denotes the number of sensors in node n_k. For simplicity, we consider that the types of sensors equipped in a node are distinct. Additionally, we denote the initial energy, storage, bandwidth, and computational capacities of each node n_k by E_k, M_k, B_k, and C_k, respectively. Further, we consider that the sensor nodes communicate their sensed data to their nearest gateway device using multihop communication.

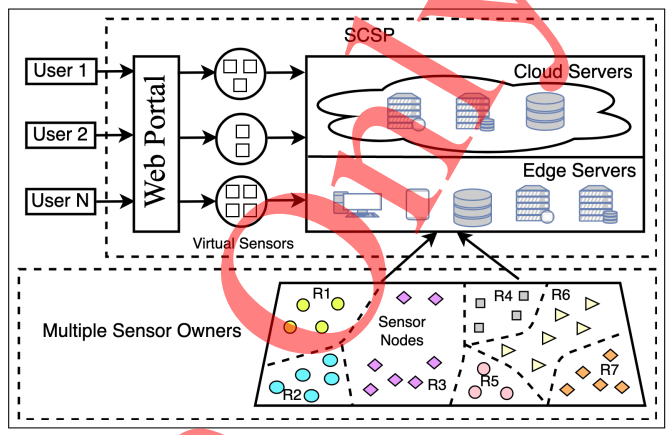

Fig. 1: Schematic Diagram of Green Sensor-Cloud

On the other hand, we consider that multiple end-users request for the service of the SCSP at each time instant. The set of service-requests handled by the SCSP at each time τ is denoted by $\mathcal{S}_\tau = \{s_1, s_2, \cdots, s_{Q_\tau}\}$, where $|\mathcal{S}_\tau| = Q_\tau$. The users specify their requirements in terms of the geographical region r_{s_q}, type of sensor t_{s_q}, data rate b_{s_q}, and maximum tolerable delay δ_{s_q}. Based on these, a Service Level Agreement (SLA) [32] is prepared among each user and the SCSP and a mutually agreed upon price p_{s_q} for the service s_q is decided. Thereafter, the SCSP propagates these requirements to the sensor-owners having compatible sensor nodes. The SCSP and the sensor-owners work together to decide the optimal set of sensor nodes to be used as source and hop nodes for provisioning the request. Let $I_{j,k}^q$ be a variable which denotes the association of a node with a service. We have,

$$I_{j,k}^q = \begin{cases} 1, & \text{if } n_k^j \text{ is assigned to serve } s_q \text{ as source node.} \\ -1, & \text{if } n_k^j \text{ is assigned to serve } s_q \text{ as hop node.} \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

While assigning services to a set of nodes, their capacity constraints must be taken into consideration. Therefore, the following conditions must be satisfied:

$$B_k \geq \sum_q |I_{j,k}^q| b_{s_q} \tag{2}$$

Additionally, the following physical constraints of each source node, where $I_{j,k}^q > 0$, need to be satisfied:

$$M_k \geq \sum_q m_{s_q} \quad \text{and} \quad C_k \geq \sum_q c_{s_q} \tag{3}$$

where m_{s_q} and c_{s_q} are the memory and computational capacity requirements of service s_q.

For the usage of their sensor nodes, the SCSP pays rent to the sensor-owners. We consider that the SCSP pays a price P^s and P^h per unit data-point for each source and hop node, respectively, to each sensor-owner. Here, $P^s > P^h$. On the

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other hand, to ensure efficient distribution of profit among the sensor-owners, we design P^h as follows —

$$P^h(n_k^j) = \psi + \omega \frac{\sum\limits_{n_k^j \in \mathcal{N}_j} \sum\limits_{q} |I_{j,k}^q|}{\sum\limits_{o^j \in \mathcal{O}} \sum\limits_{n_k^j \in \mathcal{N}_j} \sum\limits_{q} |I_{j,k}^q|} \tag{4}$$

where ψ and ω are constants; ψ denotes the minimum price to be paid for the first application to be served; and ω defines the price incremental factor. This implies that the price paid to a sensor-owner increases with the increase in the number of services assigned to his/her nodes.

Problem Scenario: To clearly understand the problem addressed in this work, we consider the scenario depicted in Figure 2. As shown in Figure 2, to serve a request s_q, there exists two compatible source nodes, A and B, belonging to owners O_1 and O_2, respectively, and both these nodes are in inactive state. The possible data transmission paths for the nodes are shown. In case of B, the path with 2 hops has nodes belonging to O_2 while that with 3 hops has nodes from O_1 and O_2 both. We observe that, although both the paths need the activation of only 1 node, these consist of a hop node with resources fully in use. Thus, both these paths are unsuitable for provisioning Se-aaS with high QoS. On the other hand, in case of A, there are two paths with adequate resources, each requiring the activation of 1 inactive node and thus, nearly equal resource consumption. However, the ownership of the nodes plays a crucial role in path selection as the two paths incur different costs to the SCSP and revenue for the owners. Thus, it is necessary to consider the effect of all the aforementioned parameters while deciding the optimal set of nodes to form the VS for s_q. Hence, in this work, we propose GS2 to address this issue.

IV. GS2: The Proposed Green Se-aaS Provisioning Scheme

The proposed scheme, GS2, is comprised of two main parts – (a) nomination of subset of nodes for VSF by each sensor-owner with compatible nodes locally, and (b) determination of the globally optimal subset among the nominated subsets of nodes for service provisioning by the SCSP. For the first part, the sensor-owners use a coalition formation game to determine the optimal coalitions of the source and hop nodes to be formed to serve a VS, while maximizing their profits. Additionally, the sensor-owners try to ensure that a minimal subset of their own nodes are activated in the network to ensure limited energy consumption, and thus, reduced maintenance costs. On the other hand, for the second part, the SCSP obtains the information of the chosen subsets of the sensor-owners and uses a utility game-based algorithm to choose the optimal one to serve the request with high profit and QoS and reduced resource consumption of the network. We discuss each of these parts in details in the following subsections.

Justification of Using Dynamic Coalition Formation and Utility Games: In sensor-cloud, the sensor-owners behave rationally and aim to maximize their individual revenue and minimize the resource consumption in terms of energy, storage, and processing, and thus, maintenance cost of their nodes. Similarly, the SCSP tries to maximize its revenue by allocating resources to serve VSs in such a way that it can serve a higher number of services with high QoS and low cost. Thus, we observe that the problem of VSF in sensor-cloud has several conflicting objectives. We argue that the aforementioned problem can be mapped to the 'bin-packing problem', which is NP-hard, considering the sensor nodes as bins and the services as the items to be packed into the bins [7]. Thus, the combinatorial optimization approaches cannot ensure a stable solution in polynomial time. Hence, to solve the problem in polynomial time, we propose a heuristic algorithm based on game theory. To model the rational behavior of sensor-owners in GS2, we use a dynamic coalition formation game, where the sensor nodes, used for forming VSs, form coalitions depending on the overlap of their data transmission path. On the other hand, we use utility game to model the optimal global path selection by the SCSP, while ensuring a trade-off between the quality of Se-aaS, network resource consumption, and the revenue of the SCSP. The proposed scheme is a combination of distributed and centralized steps and can obtain a near-optimal solution in polynomial time.

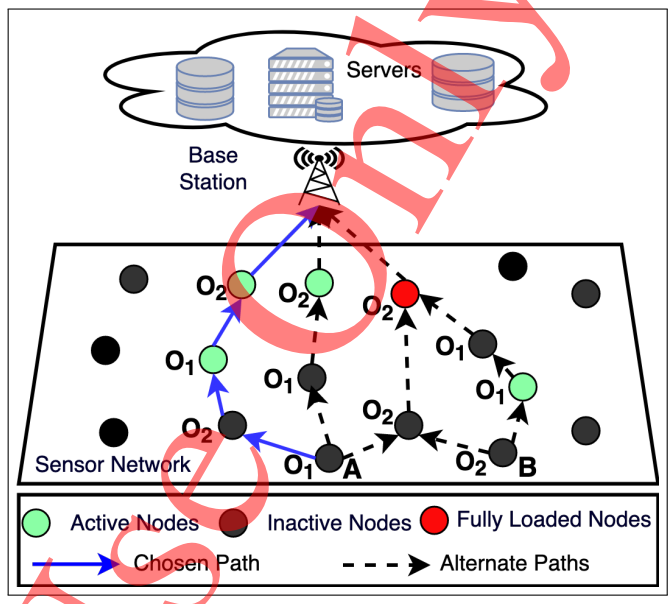

Fig. 2: Problem Scenario

A. Coalition Formation Game Formulation for Sensor-Owners

In GS2, initially, each user q specifies the service requirements, such as the geographical region r_{s_q}, type of sensor t_{s_q}, data rate b_{s_q}, and maximum tolerable delay δ_{s_q}, to the SCSP. On obtaining these specifications, the SCSP propagates the consolidated requirements to the sensor-owners with compatible sensor nodes. Based on this information, the sensor-owners select the set of nodes $\mathcal{N}_{s_q}$ which belongs the geographical region r_{s_q} and are equipped with type of sensor t_{s_q}. In the proposed scheme, GS2, the dynamic coalition formation game is formulated among this set of selected nodes. Thus, the set $\mathcal{N}_{s_q}$ of nodes act as the players and decide the optimal coalitions to form among themselves which maximizes the

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revenue of their owners and minimizes resource consumption. We discuss the details of the game as follows.

Upon selection of $\mathcal{N}_{s_q}$, each sensor-owner o_j initiates the path finding algorithm to reach the gateways for each node $n_k^j \in (\mathcal{N}_{s_q} \cap \mathcal{N}_j)$. The nodes use a reactive routing scheme to choose the neighbor nodes closest to the gateway at each step, thereby ensuring a path with the minimum possible number of hops. Here, at each step, it must be ensured that the capacity constraints of the nodes specified in Equations (2) and (3) are satisfied. Thus, at the end of the path finding phase, each source node n_k obtains an initial subset of nodes or coalition, $\mathcal{C}_k$, consisting of the initial set of intermediate nodes between n_k and gateway. Hence, initially, we get that, for each service s_q, there must be a maximum of $|\mathcal{N}_{s_q}|$ number of coalitions.

Thereafter, in GS2, the set $\mathcal{N}_{s_q}$ of source nodes decide their optimal strategies, i.e., whether to join a coalition or split from an existing one, based on a utility function. The set of available coalitions is given as $\cup_k \mathcal{C}_k \cup \mathcal{C}^{old}$, where $\mathcal{C}^{old}$ denotes the already existing coalitions currently in use for service provisioning. For each source node n_k, the strategy set $\mathcal{C}_k^{st}$ includes each coalition $\mathcal{C}_j$ for which $\exists n_i \in \mathcal{C}_j$ such that n_i is a neighbor of n_k and is closer to the gateway than n_k. Thus, the set of strategies of each player in GS2 is finite and restricted by the choice of its available paths to gateway. We discuss the utility functions of each sensor node and each coalition as follows.

Utility Function of Each Node: Each source node evaluates the utility function of each node present in each coalition in the set $\mathcal{C}_k^{st}$. The utility function $\mathcal{U}_k$ of each node n_k^j signifies the difference of the value added and the cost incurred due to node n_k^j in the coalition. Therefore, it includes the energy cost of using node n_k^j, its bandwidth usage, and the amount of revenue earned by owner o_j due to its use. Thus, the utility function $\mathcal{U}_k$ of each node n_k^j depends on the following factors and satisfies the corresponding properties:

1) *Residual Energy:* The residual energy E_k^{res} of each node n_k^j signifies its potential to be a part of a path to the gateway, as a fixed amount of energy is consumed for transmitting a message of fixed payload to the neighbor nodes. Therefore, having a high amount of residual energy ensures high payoff value, i.e., signifies a high potential node. Mathematically,

$$\frac{\partial \mathcal{U}_k}{\partial E_k^{res}} \geq 0 \tag{5}$$

2)*Available Bandwidth:* In sensor-cloud, each active node n_k^j can be part of multiple paths either as a source node or as an intermediate node. In other words, each node can serve multiple services at any given time instant. Therefore, having a high amount of available bandwidth increases the payoff of the utility function, i.e., the payoff contribution of a sensor node. Hence, from Equation (2), we get:

$$\frac{\partial \mathcal{U}_k}{\partial [\sum_q |I_{j,k}^q| b_{s_q}]} \geq 0 \tag{6}$$

3) *Activation Energy:* In case, a node is in *sleep* mode, it needs to be activated prior to service provisioning, which consumes a certain amount of activation energy. We consider that the activation energy of each node is the same and denoted as E^a. As the sensor-owners aim to reduce energy consumption, E^a has a negative effect on the utility function. Let Θ_k denotes the state of node n_k^j, where $\Theta_k = 1$ for active and $\Theta_k = 0$ for inactive. We have –

$$\frac{\partial \mathcal{U}_k}{\partial \Theta_k E^a} \leq 0 \tag{7}$$

4) *Revenue:* A high value of the revenue earned by owner o_j for unit usage of node n_k^j is desirable to the sensor-owners as they are rational entities who wish to earn maximum monetary benefit from the arrangement. Hence, the utility function varies proportionally with the revenue earned. Thus, considering the maintenance of n_k^j to be Q_k^j, we have –

$$\frac{\partial \mathcal{U}_k}{\partial [(1 + I_{j,k}^q) P^h(n_k^j) + (1 - I_{j,k}^q) P^s(n_k^j) - Q_k^j]} \geq 0 \tag{8}$$

Therefore, we define the utility function $\mathcal{U}_k$ of each active node n_k^j as follows:

$$\mathcal{U}_k = \alpha \frac{E_k^{res} - \Theta_k E^a}{E_k^{max}} + \beta \frac{B_k - \sum_q |I_{j,k}^q| b_{s_q}}{B_k} + \gamma \frac{[(1 + I_{j,k}^q) P^h(n_k^j) + (1 - I_{j,k}^q) P^s(n_k^j) - Q_k^j]}{P_{max}} \tag{9}$$

where α, β, and γ are constants and P_{max} denotes the maximum price paid to sensor-owners.

Utility Function of Each Coalition: The utility function $\mathcal{B}_l$ of each coalition $\mathcal{C}_l$ signifies the total value of the nodes in the coalition. As nodes consume energy to remain in active state, the increase in the number of active coalitions results in an increase in the network energy consumption. Therefore, in GS2, each sensor-owner aims to reduce the total number of coalitions in the network and reach an optimal configuration which maximizes his/her gain. Thus, we design the utility function $\mathcal{B}_l$ of coalition $\mathcal{C}_l$ as the sum of the utilities of the nodes present in it. Mathematically, we have –

$$\mathcal{B}_l = \sum_{n_k^j \in \mathcal{N}_{\mathcal{C}_l}} |I_{j,k}^q| \mathcal{U}_k \tag{10}$$

where $\mathcal{N}_{\mathcal{C}_l}$ denotes the set of nodes in coalition $\mathcal{C}_l$.

Objective of the Game: Using the aforementioned utility functions, we define the objective of the sensor-owners in the proposed game as follows:

$$\arg\max_{\mathcal{C}_l} \mathcal{B}_l, \quad \forall n_k^j \in \mathcal{N}_{s_q} \text{ and } \min |\mathcal{C}_l| \tag{11}$$

while satisfying the constraints in Equations (2) and (3). To obtain the aforementioned objectives, each sensor-owner o_j tries to define a preferential relation for each source node $n_k^j \in \mathcal{N}_{s_q}$ and obtain the equilibrium condition. The equilibrium condition for GS2 is defined in Definition 1.

Definition 1. *For a given service s_q, we define the equilibrium of GS2 as the optimal allocation matrix $\boldsymbol{I}^*$, where*

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$$\boldsymbol{I}^* = \begin{bmatrix} \cdots & I^{q*}_{1,k} & \cdots \\ \ddots & \vdots & \ddots \\ \cdots & I^{q*}_{|\mathcal{O},k} & \cdots \end{bmatrix} \tag{12}$$

where $I^{q*}_{1,k}$ denotes optimal allocation variable notifying if service s_q is assigned to node n^j_k. To obtain the equilibrium in GS2, the following condition needs to be satisfied:

$$\mathcal{B}^{'*}_l(n^j_k, \boldsymbol{I}_^*) \geq \mathcal{B}^{'}_l(n^j_k, \boldsymbol{I}_^*) \tag{13}$$

where $\boldsymbol{I}_^* = \boldsymbol{I}^* - \mathcal{C}_k$ and $\mathcal{B}^{'*}_l(n^j_k)$ denotes the optimal payoff value for source node $n^j_k \in \mathcal{N}_{s_q}$.

Theorem 1. *In GS2, given a fixed topology and a set of source nodes, each source node has an unique optimal path, which is considered to be equilibrium.*

Proof. We consider that for a source node $n^j_k \in \mathcal{N}_{s_q}$, the optimal sequence of intermediate nodes is $\boldsymbol{I}^*$, as defined in Definition 1. Now, we argue that there is an another sequence $\boldsymbol{I}^{+*}$ for which the following condition is true.

$$\mathcal{B}^{'*}_l(\boldsymbol{I}^{+*}) \geq \mathcal{B}^{'}_l(\boldsymbol{I}^*) \tag{14}$$

Based on Definition 1, we obtain that if the aforementioned condition is true, then the initial assumption is false. Hence, through contradiction, we prove that each source node has an unique equilibrium path using the proposed scheme, GS2. □

B. Utility Game Formulation for the SCSP

On receiving the set of selected source nodes and their corresponding paths to gateways from each sensor-owner, the SCSP uses utility game to select the optimal subset of paths, i.e., coalitions, which ensure high revenue. Hence, we model the utility function $\mathcal{S}$ of the SCSP as a trade-off between the revenue $R(\phi)$ earned by selecting a subset of coalitions ϕ and the activation energy of the new nodes. We calculate $R(\phi)$ as:

$$R(\phi) = P_{s_q} - \sum_{l \in \phi} \left[\frac{P^s}{2} \sum_{n^j_k \in \mathcal{N}_{\mathcal{C}_l}} \left(|I^q_{j,k}| + I^q_{j,k} \right) + \frac{P^h}{2} \sum_{n^j_k \in \mathcal{N}_{\mathcal{C}_l}} \left(|I^q_{j,k}| - I^q_{j,k} \right) \right] \tag{15}$$

where P_{s_q} denotes the amount of price charged by the SCSP for provisioning service s_q. Thus, we have –

$$\mathcal{S}(\phi) = x \frac{R(\phi)}{P_{s_q}} + (1-x) \frac{E^a N^a_{s_q}}{E^{max}} \tag{16}$$

where x denotes the weight factor and E^a and $N^a_{s_q}$ denote the activation energy per node and the number of new nodes being activated for ϕ, respectively. E^{max} is the maximum energy capacity of each node. Following the properties of utility game, the SCSP aims to maximize the payoff of $\mathcal{S}(\phi)$ and defines a preference relation among each pair of coalition subsets.

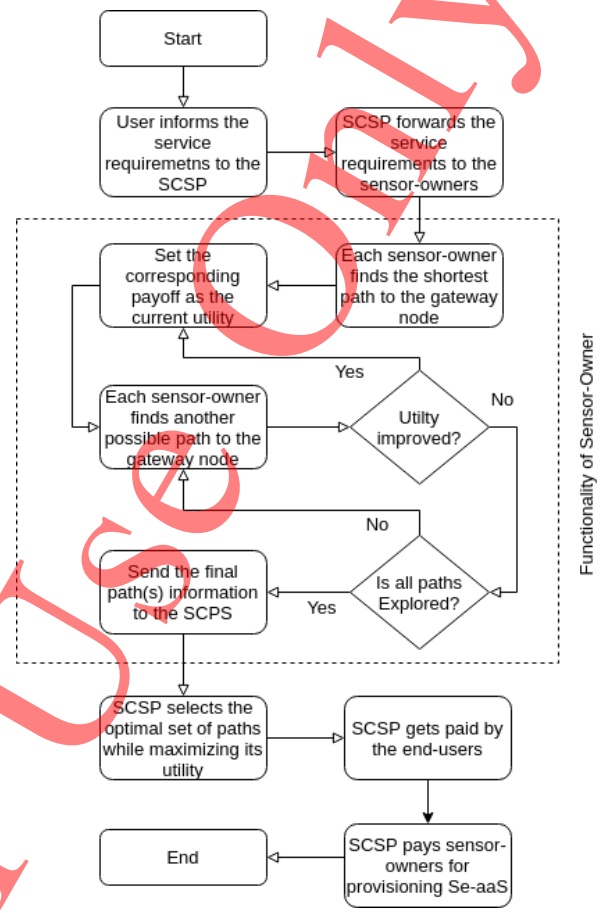

Fig. 3: Flowchart of GS2

For example, consider two subsets of coalitions ϕ_1 and ϕ_2, the SCSP follows $\phi_1 \succcurlyeq \phi_2$, if and only if $\mathcal{S}(\phi_1) \geq \mathcal{S}(\phi_2)$. Additionally, considering that $\mathbb{N}_{s_q}$ number of source nodes are required for provisioning service s_q, the SCSP needs to satisfy the following constraint, while selecting the subset ϕ:

$$\sum_{l \in \phi} \sum_{n^j_k \in \mathcal{N}_{\mathcal{C}_l}} \left(|I^q_{j,k}| + I^q_{j,k} \right) \geq 2\mathbb{N}_{s_q} \tag{17}$$

C. Proposed Workflow

Initially, in GS2, the end-users inform the SCSP about the service requirements as shown in Figure 3. On receiving the service requirements, the SCSP informs the service specification to the sensor-owners who are eligible to provision the requested service. Thereafter, each sensor-owner identifies the source nodes and evaluates their corresponding shortest path to the gateways. Thereafter, using the dynamic coalition game, each sensor-owner tries to reduce the number activated sensor nodes to ensure less resource consumption of his/her nodes and to ensure high profits for his/her-self. Towards this aim, the source nodes decide to merge into or split from each of the available coalitions on the basis of the values of the utility received, i.e., Equation (10). Finally, each sensor-owner selects the optimal set of source node and its hop nodes to the gateway to serve the request. On obtaining the choice of each sensor-owner, the SCSP evaluates the optimal subsets of nodes using

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Equation (16), while ensuring a trade-off between its profits and the energy consumption of the network.

V. Performance Evaluation

We present the performance evaluation of the proposed scheme, GS2, in comparison with the existing benchmark schemes in this section. Towards this aim, we design a simulation platform for sensor-cloud using MATLAB. The details of the simulation as well as the observed results are discussed as follows.

TABLE I: Simulation Parameters

Parameter	Value
Simulation area	10 $km\times$ 10 km
Services	[1, 5, 10] $\times$ 1,000
Types of services	5
Bandwidth per service	20-250 kbps
Maximum price per service	100 unit
Sensor owners	10
Sensor nodes per owner	100, 250, 500
Communication protocol	IEEE 802.15.4
Battery capacity of node	20 J [7]
Bandwidth capacity of node	250 $kbps$
Memory capacity of node	32 Gigabyte (GB)
Computation capacity of node	1 GB
Node signal energy consumption	50 nJ/bit [7]
Node activating energy consumption	30 nJ/bit [7]
Sensing energy	10 $nJ/bit\text{-}m^2$ [7]
x	0.5

A. Simulation Parameters

We considered a single SCSP with 10 registered sensor-owners, each having deployed varying number of nodes, randomly, over a 10$\times$10 km^2 area. Each nodes is equipped with 3-5 types of sensors which are chosen randomly. Considering that the nodes use the IEEE 802.15.4 Zigbee protocol for communication, the maximum supported data-rate and range of communication are 250 $kbps$ and 100 m, respectively. On the other hand, the requests of the users arrive to the SCSP gradually and have randomly generated requirements. In this case, we consider that, for serving the requests, the chosen source nodes compute their utility for the shortest path which satisfies the resource capacity constraints. The detailed simulation parameters are listed in Table I. It is noteworthy that the simulations were designed considering the typical range of values of the different parameters chosen in the existing work of Misra and Chakraborty [7].

B. Benchmarks

We compared the performance of the proposed scheme with two existing benchmarks – QADMAP [7] and IoT Services Allocation at the Edge (ISA) [21]. In QADMAP, Misra *et al.* proposed a QoS-aware scheme for mapping of physical sensors to VSs in sensor-cloud. The authors used p-dispersion to choose the set of compatible nodes and used an evolutionary game to decide the optimal mappings, while ensuring even load distributions. However, in this work, the authors only considered the load on the source nodes and not the intermediate hop nodes.

On the other hand, in ISA, Amadeo *et al.* proposed a NDN-based framework, IoT-NCN, and proposed a service allocation strategy using an ILP-based formulation. The authors considered a simple additive weighting of the delay cost and the processing cost to generate the objective function and proposed a distributed algorithm to solve the same. However, in this work, the authors neither considered the economic aspects of IoT services nor the energy consumption due to multihop wireless communication.

C. Results and Discussions

We discuss the results of performance analysis of the three schemes in terms of various system parameters as follows.

*1) **Energy consumption of nodes**:* The energy consumption of the nodes is the sum of their activation energy, energy expended to remain in active state, and the energy consumed due to transmission and reception of sensed data. Additionally, the source nodes consume energy for data sensing. To realize a green sensor-cloud, we aim to reduce the overall energy consumption of the network in this work. From Figures 4, we observe the variation of energy consumption for the three schemes, with varying number of nodes and varying number of services. We observe that the energy consumption is at least 15.92% and 24.18% less using GS2 compared to QADMAP and ISA, respectively, and it increases with the increase in the number of services. This is because, firstly, both GS2 and QADMAP are designed with the aim of reducing energy consumption, unlike ISA. Additionally, unlike QADMAP, GS2 decides the physical sensor nodes to be used for forming a VS, while considering their path to the gateway device and gives preference to those paths which have a high number of active nodes with remaining capacity. This helps to reduce the energy consumption even further, while provisioning green Se-aaS. Additionally, we observe that with the increase in the number of nodes in the network, the energy consumption decreases, which can be attributed to the increase in the availability of shorter paths to the gateway.

*2) **Network lifetime**:* We calculate the network lifetime as the time duration elapsed from the network deployment to the time when the last node dies. The network lifetime varies inversely with the energy consumption of the network and is a measure of how long the SCSP remains in service. Thus, a high value of network lifetime signifies reduced resource consumption and a greener sensor-cloud. Figure 5 shows that the variation of the network lifetime follows nearly the inverse pattern of the energy consumption and is higher by 29.74% and 30.1% using GS2 compared to QADMAP and ISA, respectively. As the number of services increases, the network lifetime decreases, whereas, as the number of nodes increases, the network lifetime increases using the three schemes.

*3) **Service failures**:* We consider that a service failure occurs when the energy of source node gets exhausted or there is no available path to gateways due to insufficient energy of

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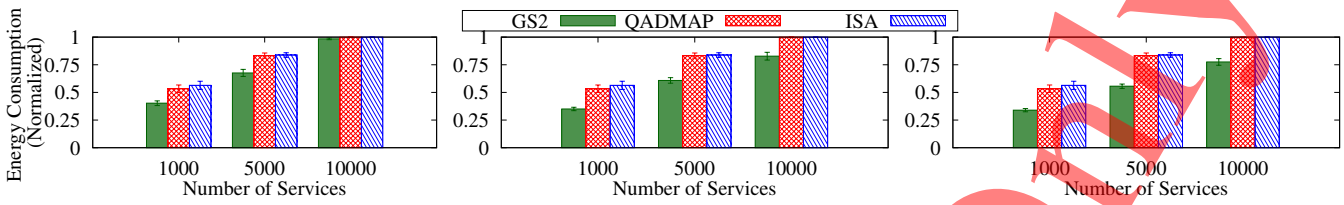
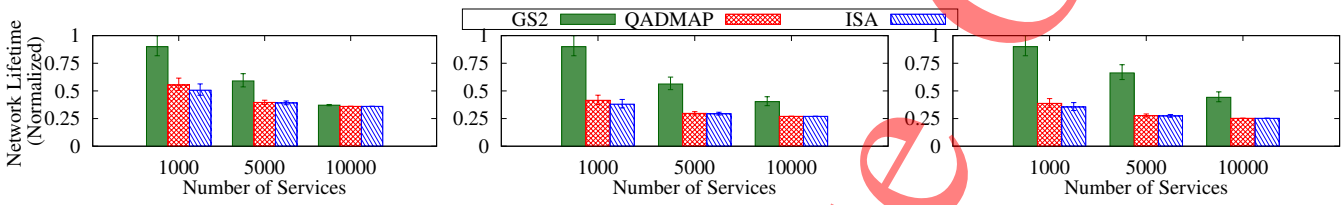


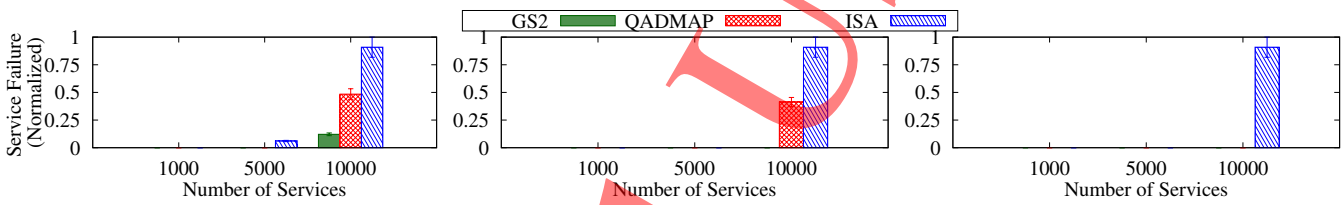
(a) Sensor Nodes/Sensor-Owner = 100 (b) Sensor Nodes/Sensor-Owner = 250 (c) Sensor Nodes/Sensor-Owner = 500

Fig. 4: Energy Consumption of the Nodes



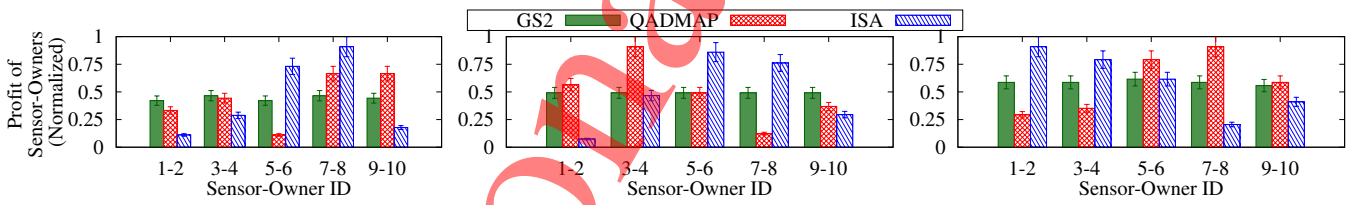
(a) Sensor Nodes/Sensor-Owner = 100 (b) Sensor Nodes/Sensor-Owner = 250 (c) Sensor Nodes/Sensor-Owner = 500

Fig. 5: Network Lifetime



(a) Sensor Nodes/Sensor-Owner = 100 (b) Sensor Nodes/Sensor-Owner = 250 (c) Sensor Nodes/Sensor-Owner = 500

Fig. 6: Service Failure



(a) Number of Services = 1,000 (b) Number of Services = 5,000 (c) Number of Services = 10,000

Fig. 7: Profit of Sensor-Owners

the hop nodes. In addition, we also consider those services which cannot be served due to the inadequacy of sensor nodes with utilized capacity. We calculate this parameter in the form of a ratio with the total number of services requested by the users. Service failures signify QoS violations, user dissatisfaction, as well as decrease in the market reputation of the SCSP. Figure 6 depicts the service failures using the three schemes. We observe that the percentage of service failures are lowest, i.e., around 0.18-0.22%, using GS2, compared to 75.01% using QADMAP and 86.67% using ISA. In addition to energy, the service failures also depend on bandwidth capacity, which is not considered in the other two schemes. Further, in ISA, the nodes closer to the gateway are always prioritized and thus, are exhausted of resources sooner than those which are far away, thereby increasing the service failures even further. As the number of nodes increases, the service failures decrease using all the three schemes, as higher number of nodes implies higher number of possible sources and paths to the gateway.

*4) **Profit of Sensor owners**:* The profit earned by the sensor owners depends on the usage of their nodes and the cost incurred due to maintenance. We observe the variation in the profit earned by the sensor-owners using the three schemes in Figure 7. We yield that the profit earned by the sensor-owners is nearly evenly distributed using GS2, whereas using QADMAP and ISA, it varies randomly. This is because, in GS2, the VS composition takes into consideration the profit distribution of the sensor-owners, while choosing both the source and hop nodes, unlike QADMAP in which it is considered only when deciding the source nodes and ISA in which it is not considered at all. Further, due to the optimized usage of the sensor nodes and activation of less number of nodes, the sensor-owners incur lower maintenance cost but same amount of revenue from service provisioning, thereby increasing the overall profit. Thus, the profit is higher by at least 21.05% and 38.09% using GS2 than using QADMAP and ISA, respectively. With the increase in services, although

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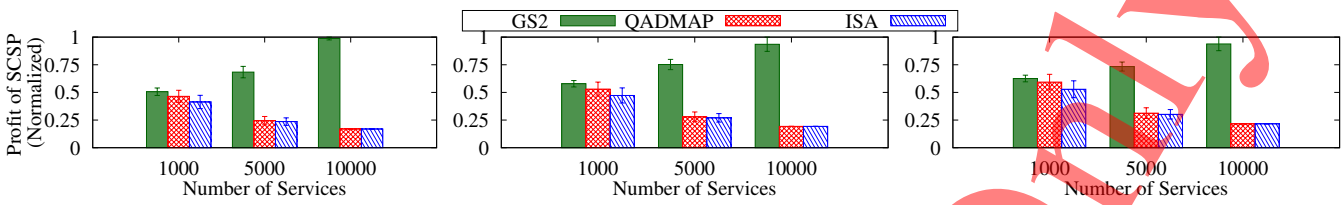

(a) Sensor Nodes/Sensor-Owner = 100 (b) Sensor Nodes/Sensor-Owner = 250 (c) Sensor Nodes/Sensor-Owner = 500

Fig. 8: Profit of SCSP

the usage of the nodes increase, so does the revenue earned by service provisioning as the same node can be used to serve a higher number of services. Thus, the net profit of the sensor-owners increase. Further, the number of nodes do not have any significant effect on the profit earned by the sensor-owners. However, with increasing nodes, the number of services that can be served increases, which brings about a positive change in the net profit of the sensor-owners.

*5) **Profit of SCSP**:* The profit earned by the SCSP depends on the number of services served successfully, the resources utilization, and the revenue paid to the sensor-owners. Figure 8 depicts the variation in the profit of the SCSP with the change in the number of services and sensor nodes, using the three schemes. We observe that, using GS2, the profit of the SCSP increases with the increase in number of services. This is due to the fact that, with the increase in the number of services, the number of active nodes does not increase significantly. Thus, the SCSP earns higher revenue while incurring a lower cost. From Figure 8, we yield that the profit of the SCSP increases by at least 8.33-18.33% using GS2, than using QADMAP and ISA. Moreover, we observe that the profit of the SCSP decreases using the other two schemes – QADMAP and ISA, as the SCSP needs to pay a minimum price to the owners of each activated node. On the other hand, we observe that, with increase in the number of nodes, there is only a slight change in the profit of the SCSP as the number of nodes does not have a significant effect on the service provisioning in GS2.

Hence, based on these aforementioned performance metrics, we claim that the proposed scheme, GS2, outperforms the existing schemes – QADMAP and ISA.

VI. Conclusion

In this work, we addressed the problem of VSF in green sensor-cloud. We used a game theoretic approach to model the interactions of the sensor owners and the SCSP. Based on this, we proposed GS2, a hybrid algorithm to decide the optimal source nodes and their corresponding data transmission path to the sink, while ensuring reduced energy consumption and service provisioning cost, and high QoS of Se-aaS. Specifically, the sensor owners choose a set of suitable source nodes and their corresponding set of paths to the gateway. Thereafter, using coalition game, the sensor owners decide the optimal path for each node which gives maximum utility. On obtaining the set of paths from the sensor-owners, the SCSP uses utility theory to choose the optimal set of source and hop nodes to be used to form the VS corresponding to a request, while considering its profit and the resource consumption of the network. We evaluate the performance of our proposed scheme GS2, through simulations, and show that it outperforms the existing benchmarks significantly.

This work can be extended in the future while considering the presence of mobile nodes, as the dynamic nature of node positions directly impact the data transmission paths. It can also be extended to study the effect of selfish behavior of sensor owners on the energy consumption of the network.

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